

Apollo 11 landing-time reconciliation

Demonstrate provenance, exact locators, temporal uncertainty, rights review, contradiction handling, and deterministic export using metadata-only official records.

SOURCES 3 reviewed public locators	CLAIMS 4 source-linked propositions	CONFLICTS 1 preserved contradiction ledgers	PRESERVED BODIES 0 rights-permitted remote bodies
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STATUS: WORKING / RESEARCH CUTOFF: 2026-07-16T20:30:00Z / SCHEMA: 2.0.0

This analysis is not a finding of guilt, certification, metric-validation statement, court-admissibility opinion, or public-release approval.

Executive summary

ESTABLISHED IN PACKAGE

Two official NASA records report that Eagle landed on the lunar surface; their exact recorded mission elapsed times differ by 3.1 seconds.

ESTABLISHED IN PACKAGE

The Apollo 11 Mission Report records landing on the surface at mission elapsed time 102:45:39.9.

ESTABLISHED IN PACKAGE

NASA's Record of Lunar Events lists lunar landing at mission elapsed time 102:45:43 and 20:17:43 GMT on July 20, 1969.

OPEN EVIDENCE NEED

Which timing and clock-reference conventions explain the 3.1-second difference between the two official records?

Method and evidence profile

INCLUSION RULE Public records and specifically reviewed metadata-only fixtures relevant to the defined investigation scope.	EXCLUSION RULE No crawler execution, authenticated access, raw third-party body retention, publication, or unsupported identity, guilt, trajectory, or metric inference.	SOURCE HIERARCHY preserved primary evidence > official record > independent reporting > attributed analysis > unresolved assertion
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CLAIM EVIDENCE STATES
authenticated official record 4

SOURCE AUTHORITY CLASSES
primary official 2
published secondary 1

Chronology

NO.	SOURCE TIME	EVENT	SOURCE-BOUNDED DESCRIPTION
01	102:45:39.9 / 102:45:43 / 20:17:43 GMT 20 July 1969	Lunar Module landing	Official NASA records agree that Eagle landed but list mission elapsed times that differ by 3.1 seconds.
02	109:24:15 / 02:56:15 GMT 21 July 1969	Armstrong on the lunar surface	NASA's Record of Lunar Events lists Armstrong on the lunar surface at 109:24:15 mission elapsed time and 02:56:15 GMT on July 21, 1969.

Contradiction register

Touchdown mission-time discrepancy

OPEN

The mission report lists 102:45:39.9 while the event record lists 102:45:43. The records differ by 3.1 seconds but agree that the landing occurred.

Different definitions of first surface contact versus recorded landing event
Clock-source or synchronization differences
Later administrative rounding or transcription

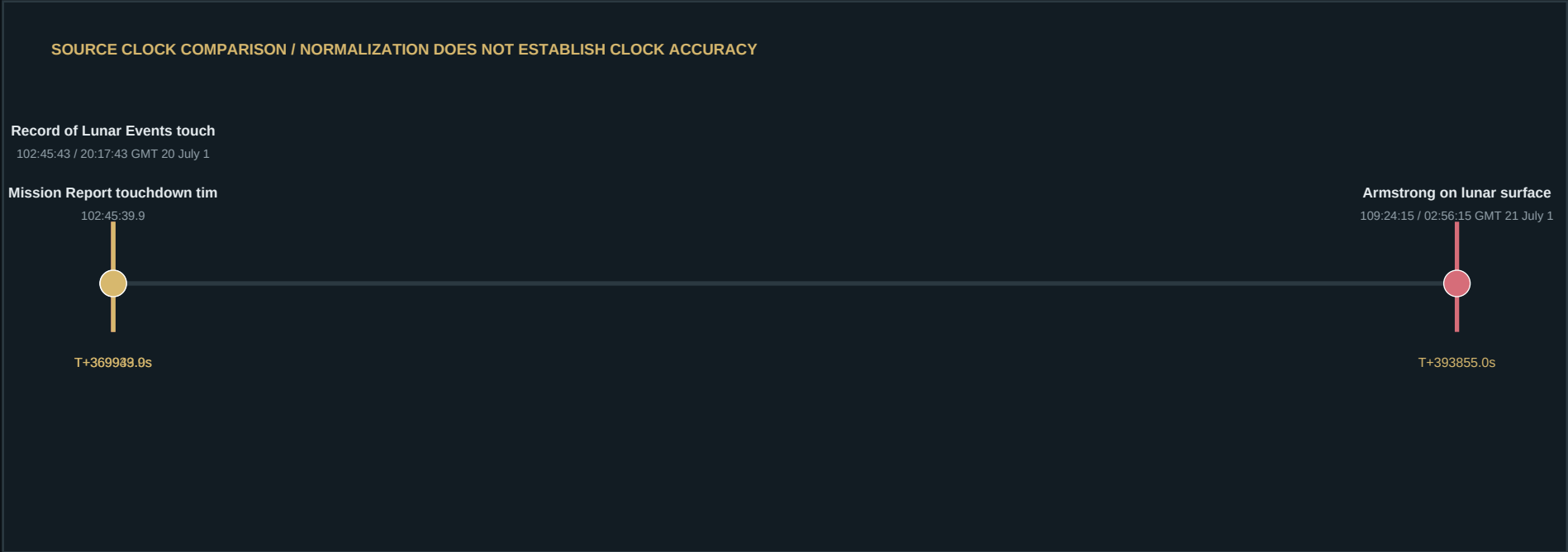
Spatial and integrity gates

ARTIFACT	FRAME	METRIC	MEASUREMENT	VALIDATION
Touchdown time comparison marker	LOCAL_UNITLESS	NO	DISABLED	not_metric

STRUCTURAL VALIDATION	EVIDENCE AUTHENTICATION	PRESERVED SOURCE BODIES
passed_v2_audit	limited_metadata_only	0 of 3

Internal consistency does not authenticate remote source content. Measurements remain disabled for non-metric case scenes. Gaussian-splat products are visualization derivatives unless independently validated.

Labeled reconstruction view



This diagram is generated from the package's relational scene or normalized timing anchors. It contains no copied source pixels and does not add metric, geodetic, trajectory, identity, or legal conclusions.

Source appendix

ID	SOURCE	LOCATOR	PRESERVATION
S001	Apollo 11 Mission Report National Aeronautics and Space Administration https://www.nasa.gov/wp-content/uploads/static/apollo50th/pdf/A11_MissionReport.pdf	section: 5.4 Landing Dynamics; page: Report page 5-14	metadata_only
S002	Apollo 11 Record of Lunar Events National Aeronautics and Space Administration https://www.nasa.gov/wp-content/uploads/static/history/ap11ann/ap11events.html	row: Lunar Landing Time; row: Armstrong on Lunar Surface	metadata_only
S003	NASA Images and Media Usage Guidelines National Aeronautics and Space Administration https://www.nasa.gov/nasa-brand-center/images-and-media/	section: NASA Images and Media Usage Guidelines	metadata_only